MIL-R-85535(AS)
AMENDMENT 2
26 August 1986
SUPERSEDING
AMENDMENT 1
14 DEC 1983

MILITARY SPECIFICATION

RECORDER-REPRODUCER SET, DIGITAL DATA, AN/ASH-33

This amendment forms a part of Military Specification MIL-R-85535(AS) dated 8 June 1983, and is approved for use by the Naval Air Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.1, line 15: Delete "AN/AYK-114(V)" and substitute: "AN/AYK-14(V)."

PAGE 3

2.1.1, After MIL-R-5757 add:

"MIL-E-6051 Electromagnetic Compatibility Requirements Systems"

The attached insertable replacement pages listed below are replacements for stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

Replacement Page	Page Replaced
47	Reprinted without change
48	48
49	49
50	50
62A	new page
62B	new page
83	83
84	Reprinted without change
85	85
86	Reprinted without change
97	Reprinted without change
98	98
99	99
100	Reprinted without change

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PAGE 4

2.1.1, After MIL-H-46855 add:

"MIL-P-55110 Printed Wiring Boards"

2.1.1, After MIL-C-83421 add:

"MIL-C-85485 Cable, Electric, Filterline, Radio Frequency Absorptive

PAGE 5

2.1.1, After MIL-STD-210 add:

"MIL-STD-275 Printed Wiring for Electronic Equipment"

2.1.1, After MIL-STD-810 add:

"MIL-STD-882 System Safety Program Requirements"

PAGE 6

2.1.1, Delete "MIL-STD-1679 Weapons System Software Development" and substitute:

"DoD-STD-1686 Electrostatic Discharge Control Program for Protection of Electrical and Electronic Parts, Assemblies and Equipment"

2.1.1, After MIL-STD-2084 add:

"DoD-STD-2167 Defense System Software Development"

2.1.1, After MIL-HDBK-217 add:

"DoD-HDBK-263 Electrostatic Discharge Control Handbook for Protection of Electrical Electronic Parts, Assemblies and Equipment (Excluding Electrically Initiated Explosive Devices)"

2.1.1, After MIL-HDBK-472 add:

"NAVAL MATERIAL COMMAND

NAVMAT P4855-1 Navy Power Supply Reliability Design and Manufacturing Guidelines

NAVMAT P9492 Navy Manufacturing Screening Program

BUREAU OF MEDICINE AND SURGERY (BUMED)

BUMED P5055 Radiation Health Protection Manual"

2.1.2, Line 5: Delete: "DEPARTMENT OF THE NAVY."

2.1.2, Line 6: Delete: "SECNAVINST 3560.1 Tactical Data Systems Documentation Standards"

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PAGE 7

Below PUBLICATIONS add:

NAVAL AIR SYSTEMS COMMAND (NAVAIR)

WS-6536 Procedures and Requirements for Preparation and Soldering of Electrical Connectors"

PAGE 9

3.2.4.1, Line 2: add:

"NAVMAT P9492 and" between "shall be screened to" and "MIL-STD-883"

3.2.4.1, Lines 5-7: Delete Sentence:

"The functional testing shall be done at room ambient, -55°C and +125°C temperatures at a facility other than the parts manufacturer's." and substitute: "Functional testing shall include Particle Impact Noise (PIN) tests and electrical tests that check the electrical response of parts at low (-55°C), medium (room ambient) and high (+125°C) temperatures. Functional testing shall be done at a facility other than the parts manufacturer's."

PAGE 13

3.2.5, After last sentence add:

"Rigid printed wiring and printed wiring boards on single-sided, double-sided and milti-layered printed wiring shall conform to MIL-STD-275 and MIL-P-55110."

PAGE 16

3.3.3.2, Delete in its entirety and substitute:

"3.3.3.2 Electrical interconnections. The equipment shall be designed to provide specified performance when interconnected with wiring in accordance with MIL-W-5088 and MIL-E-6051. General purpose wiring shall conform to MIL-W-5088, MIL-C-85485 and WS-6536."

PAGE 23

3.3.6.2.2, After paragraph 3.3.6.2.2. add new paragraph 3.3.6.2.3:

"3.3.6.2.3 Protection against Electrostatic Discharge (ESD). The parts, SRAs, and items shall be designed so the microelectric circuits are not susceptable to damage or upset from ESD due to handling by operating or maintenance personnel. DoD-HDBK-263 and DoD-STD-1686 shall be used as guides for providing this protection."

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3.3.12.3, After paragraph 3.3.12.3 add new paragraph 3.3.12.4:

"3.3.12.4 Power supply design. Low voltage power supplies, integral or independent shall be designed to the following requirements using NAVMAT P4855-1 as a guide.

a. The MTBF of each power supply shall be no less that the allotted value required to meet the overall system MTBF, not in any case less than 40000 hours.

b. Power supply power density (output power divided by power supply envelope volume, including electromagnetic interference where required) shall be not greater than three watts per cubic inch, using standard design and construction techniques, as defined in NAVMAT P4855-1.

c. Junction temperature of semiconductor devices (diodes, transistors, hybrids and monolithic integrated circuits) shall be not greater than +100°C.

d. Power supplies designed to be repairable shall be repairable at intermediate and depot maintenance levels."

3.3.20, Delete in its entirety and substitute:

"3.3.20 Safety. The contractor shall conduct a system safety program in accordance with the applicable requirements of MIL-STD-882 (See 6.2.2). Primary emphasis of the safety program should be on the hazards associated with the interrelationships between the operating and maintenance personnel, the air vehicle, the various avionics devices, and the avionics system designed to integrate this equipment into the air vehicle. For example: The system shall be analyzed to determine if unplanned or inadvertent commands are generated, critical controls shall be designed and located to preclude inadvertent operation, computer programs shall include logic to prevent the system from entering into unsafe or dangerous mode of operation. The system and its components shall be designed and the CFE developed for safety in accordance with MIL-E-5400.

3.3.20.1 X-Radiation exposure. X-radiation producing equipment shall be designed to limit the personnel dosage rate at any readily accessible point, measured two inches from the outer surface, to a 0.5 millirem per hour with external shielding removed or 500 millirem per year (BIMED NAVMED P5055). This requirement applies to the areas normally inhabited by personnel during installation, calibration, maintenance, and operation of equipment.

3.3.20.2 Electromagnetic radiation exposure. The system design criteria shall include provisions for protection of personnel from RF electromagnetic, electrostatic and shock hazzards in accordance with requirement 1 of MIL-STD-454. Where possible, protection provisions shall be designed into associated subsystem/equipments. When protection by design is not technically feasible, adequate safety precautions shall be provided.

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3.3.20.3 Toxicity. Personnel exposure to toxic air contaminants during system operation or maintenance shall be evaluated and minimized. In addition, any malfunction of the electrical power and distribution system, avionics or other electrical system component in the form of but not limited to electrical fires and battery chemical releases shall not release into the cockpit or cabin area any gases, fumes, vapors or particulates which could incapacitate the crew and subsequently result in inability to successfully perform an emergency landing or ditching and egress. These gases, fumes, vapors and particulates shall be positively prevented from leaking back into the cockpit or cabin area from when they are purged.

3.3.20.4 Electronic equipment hazards. All newly designed electronic equipments and modified portions of existing electronic equipment shall be in accordance with MIL-STD-454 requirement 1."

PAGE 40

3.4.1.3.1, last line: Delete: "3.5.2.2.2." and substitute: "3.5.1.3.2.1."

3.4.1.3.2, last line: Delete: "3.5.1.3.2.1." and substitute: "3.5.1.3.2.2."

3.4.1.3.3, last line: Delete: "3.5.1.3.2.2." and substitute: "3.5.1.3.2.3."

PAGE 42

3.4.4, last line: Delete:

"SECNAVINST 3560.1 and MIL-STD-1679" and substitute: "DoD-STD-2167"

PAGE 43

TABLE V, Description column, line 17: Delete:

"Figures 10A, 10B, and 11" and substitute: "Figures 10, 11, and 12."

PAGE 46

3.5.1.1. Add: "h. Control each tape unit's tape motion, in DMTU Auto Mode."

PAGE 48

Figure 10: Delete and substitute attached FIGURE 10.

PAGE 49

Figure 11: Delete and substitute attached FIGURE 11.

PAGE 50

Figure 12: Delete and substitute attached FIGURE 12.

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PAGE 52

TABLE IV, Type Device column, lines 18 through 23: Delete:

"Mon" and substitute "Alt."

PAGE 54

3.5.1.3.2.2, line 11: Delete: "3.5.2.2.3" and substitute "AD 1404."

3.5.1.3.2.2, line 13: Delete: "3.5.2.2.3.3.1" and substitute "AD 1404."

3.5.1.3.2.2.1, line 6: Delete: "X" and substitute "X-1."

PAGE 55

TABLE VII, Add vertical line between "Exception" and "Display" Columns.

TABLE VII, Exception column, line 6: Delete: "Out" and substitute "Out (TO)."

TABLE VII, Normal column, line 15: Delete: "DC" and substitute "DW."

PAGE 56

TABLE VII, NOTES: After "CIW" add

"CRS = Channel Reset
SC = System Controller
TO = Times Out"

TABLE VII, NOTES, lines 11 and 12: Delete: "The IDS" and substitute "AD 1404."

PAGE 57

TABLE VIII, Title: Delete: "ASP" and substitute "SASP."

TABLE VIII, Exception column, line 6: Delete: "Out" and substitute "Out (TO)."

PAGE 58

TABLE VIII, Title: Delete: "ASP" and substitute "SASP."

TABLE VIII, Notes: After "DW" add

"CRS = Channel Reset
SC = System Controller
TO = Times Out"

TABLE VIII, NOTES, lines 10 and 11: Delete: "The IDS: and substitute "AD 1404."

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TABLE IX, Exception column, line 6: Delete: "Out" and substitute "Out (TO)."

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TABLE IX, Notes: After "IPL" add

"CRS = Channel Reset
SC = System Controller"

TABLE IX, NOTES, line 7: Delete: "Time Out" and substitute "Times Out."

TABLE IX, NOTES, lines 13 and 14: Delete: "The IDS" and substitute "AD 1404."

PAGE 61

TABLE X, Exception column, line 6: Delete: "Out" and substitute "Out (TO)."

PAGE 62

TABLE X, NOTES: After "IPL" add

"CRS = Channel Reset
SC = System Controller"

TABLE X, line 8: Delete: "Time Out" and substitute "Times Out."

TABLE X, NOTES, lines 13 and 14: Delete: "The IDS" and substitute "AD 1404."

3.5.1.3.2.2.1, line 5: Delete: "four" and substitute "five."

3.5.1.3.2.2.1, line 9: Delete: "X" and substitute "X-1."

Add Table X-1 as page 62A.

Add Table X-1 (Continued) as page 62B.

PAGE 63

3.5.1.3.2.2.1: Add: "e. System Loader - Table X-1."

3.5.1.3.2.2.1, last line: Delete: "and X" and substitute "X and X-1."

PAGE 64

3.5.1.3.2.2.1.5, after paragraph 3.5.1.3.2.2.1.5, and new paragraph 3.5.1.3.2.2.1.6:

"3.5.1.3.2.2.1.6 System Loader (SLDR) load option. When implemented in the hardware, initiating the system loader load option shall initiate the command sequence specified in Table X-1. Receipt of the CIW sequences and DW sequences shown in the table shall cause the corresponding messages to be displayed to the operator."

PAGE 83

Figure 18: Delete and substitute attached FIGURE 18.

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Figure 19: Delete and substitute attached FIGURE 19.

PAGE 87

TABLE XI, Control or Indicator column, line 9: Delete:

"IND" and substitute "LAMP."

PAGE 89

3.5.3.1.3.1.2.3.2.f, line 1: Delete: "Reverse" and substitute "Rewind."

PAGE 92

3.5.3.1.3.2.5, line 2: Delete:

"have no more than one permanent error per 10^6 bits." and substitute "be free of permanent errors over the useable tape length."

PAGE 94

TABLE XII, Requirement Paragraph column, line 3: Delete: "3.3."

TABLE XII, Requirement Paragraph column, Line 8: Delete:

"3.3.13.7" and substitute: "3.3.13."

TABLE XII, Requirement Paragraph column, line 19: Delete:

"3.3.10" and substitute "3.3.10.4."

TABLE XII, Requirement Paragraph column, last line: Delete:

"3.4.3.5" and substitute "3.4.4, 3.5."

PAGE 95

TABLE XIII, Requirement Paragraph column, line 3: Delete "3.3."

4.5.1.3, line 5: Delete: "+85°C" and substitute "+55°C."

PAGE 98

FIGURE 23: Delete and substitute attached FIGURE 23.

PAGE 99

FIGURE 24: Delete and substitute attached FIGURE 24.

PAGE 100

4.5.1.3.1: Delete subparagraphs b through h and substitute:

"b. High operating temperature: +55°C (with dummy cartridge)

c. High operating temperature: +50°C (with operational tape)

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d. Low operating temperature: -25°C (with dummy cartridge)

e. Low operating temperature: 0°C (with operational tape)

f. Chamber temperature rate: 5°C per minute (minimum)"

PAGE 104

4.5.6.2, line 3: Delete: "Method 9" and substitute "Method 4."

PAGE 105

4.5.2.5, line 3: Delete: "Method 9" and substitute "Method 4."

PAGE 106

4.7, last line: Delete: "4.9.5" and substitute "4.8."

PAGE 111

6.2.2, Add new data requirements after 3.3.19:

Paragraph No.	Data Requirement Type	Applicable DID No.	Options
"3.3.20	System Safety Program Plan	DI-SAFT-80100	-
	System Safety Report"	DI-SAFT-80104	-

PAGE 112

6.2.2, Delete all items under paragraph 3.4.4 and add:

Paragraph No.	Data Requirement Type	Applicable DID No.	Options
"3.4.4	Interface Design Document	DI-MCCR-80027	-
	Software Development Plan	DI-MCCR-80030	-
	Data Base Design Document	DI-MCCR-80028	-
	Software Requirements Specification"	DI-MCCR-80025	-

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6.2.2, Delete all items under paragraph 4.5.7 and add:

Paragraph No.	Data Requirement Type	Applicable DID No.	Options
4.5.7	"Computer Resources Integrated Support Document	DI-MCCR-80024	-
	Software Quality Evaluation Plan	DI-MCCR-80010	-
	Computer System Operator's Manual	DI-MCCR-80018	-
	Software Test Procedures	DI-MCCR-80016	-
	Software Test Report	DI-MCCR-80017	-

6.2.2, paragraph 4.9.2, line 2: Delete:

"Assurance DI-R-2174" and substitute "Evaluation DI-MCCR-80010"

Preparing activity
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